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ClinNetDx: A Novel Explainable AI Framework for Acute Lymphoblastic Leukemia Detection Using Multi-modal Clinical Data

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Abstract - Early identification of Acute Lymphoblastic Leukemia (ALL) is vital to respond on time; yet, presently existing diagnostic procedures still depend on laboratory knowledge, manual interpretation and many supplementary exams. This paper introduces ClinNetDx, a new hybrid deep learning model for Acute Lymphoblastic Leukaemia prediction that combines a Convolutional Neural Network (CNN) and a Deep Neural Network (DNN) and employs structured blood test data and (optional) medical images to support good health and well-being. The proposed methodology includes a tabular-to-spatial transformation that converts numerical clinical measures into 2D representations, enabling a CNN to learn spatial correlations among features from structured medical data. We use advanced pre-processing methods (normalization, imputation, SMOTE) that are robust to class imbalance and missing variables. For interpretability, we use dual explainability: SHAP values to explain feature contributions for numerical data and Grad-CAM visualizations to generate localized heatmaps for image inputs. To make it accessible in a clinical environment, the system is built as a Django-powered web application that provides on-the-fly predictions and an intuitive presentation of findings. The

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design is flexible and deployed as a RESTful API for interface with medical information systems, mHealth apps and cloud platforms. To demonstrate the promising predictive and generalizing power of the proposed system, it was evaluated on publicly available multi-modal haematological datasets comprising 5,000 structured CBC records, 2,500 ALL smear images, and 2,000 synchronized multi-modal samples using k-fold cross-validation, with confusion matrices and ROC-AUC reported. Experimental assessment shows that ClinNetDx achieves 98.1% classification accuracy, outperforming conventional machine learning and unimodal deep learning models while maintaining strong precision, recall, and ROC-AUC performance. The proposed system combines accurate multi-modal prediction with interpretable decision support, making it a potential option for computer-assisted early diagnosis of Acute Lymphoblastic Leukaemia and for future clinical decision-support applications.

Keywords: Acute Lymphoblastic Leukemia, Early Diagnosis, Deep Learning, Convolutional Neural Network, SHAP Explainability, Good health and well-being.

1. Introduction

Acute lymphoblastic leukaemia (ALL) is a malignant haematological disorder defined by the unregulated proliferation of immature lymphoid progenitor cells within the bone marrow and peripheral blood. It is one of the most common types of leukaemia, especially in children, but it also occurs in adults. Early diagnosis is important as any delay in treatment greatly reduces the chances of survival.

Mathematical modelling has elucidated analogous dynamics in epidemic diseases, showing how rapid proliferation and systemic effects demand early intervention to achieve better outcomes [1–4]. As with other diseases, prolonged treatment will significantly worsen prognosis due to the high proliferation rate and systemic effects of malignant cells, and early diagnosis remains the most important factor in improving patient survival [5, 6]. The standard diagnostic workup for ALL includes a complete blood count (CBC, number and types of blood cells), stained peripheral blood smear examination under the microscope, flow cytometry subtyping, and invasive techniques such as bone marrow aspiration. Even if clinically proven, these methods are laboratory-based, skill-based, and time-consuming. This imaging strategy may not be available in many health care settings, especially in rural or resource-poor settings, thereby limiting diagnostic pathways and resulting in a larger proportion of individuals presenting with advanced medical conditions [7].

Recent machine learning methods have reported promising results in predicting Acute Lymphoblastic Leukaemia from blood test markers and peripheral smear images [8–11]. CNN-based models have achieved high accuracy in classifying leukaemia from microscopic blood images [12]. Also, optimized SVM methods have improved the identification of chronic lymphoblastic leukaemia [13], and XGBoost is employed for multi-cancer classification based on ctDNA mutations [14]. Other CNN implementations using MobileNet [15], VGG16 transfer

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learning [16], and conference-demonstrated models [17-19] demonstrate the versatility of deep learning for haematological malignancies. However, many existing models are not interpretable, struggle with imbalanced datasets, and require intricate frameworks, thereby limiting their clinical deployment [20, 21]. Modern immediate care solutions and dimensionality-reduction techniques are possible options for resource-limited settings [22]; however, the combination of numerical blood data and imaging via spatial transformation has been relatively unexplored.

AI has set a precedent with its strong, sensitive, and precise role in medical diagnostics, where other roles are limited. Deep learning models such as CNNs have achieved state-of-the-art results in clinical image analysis across various imaging modalities, including radiography, histopathology, dermatology, and ophthalmology. Furthermore, when examining structured clinical data, such as comorbidity ratings, patient age, and test results, DNNs have demonstrated excellent predictive accuracy. Our models outperformed traditional machine learning techniques on large, carefully selected datasets. Nevertheless, most deep learning methods currently used for hematologic cancer detection are unimodal. While some rely mostly on organized data in CBCs or EHRs, others emphasize on smear scans. By limiting model generalization, such a silo-based method restricts the robustness and depth of diagnosis [23]. Conversely, most academic models optimize for internal performance metrics like accuracy or AUC, but not for key medical needs like explainability, interpretability, or readiness for deployment, which in the end determine whether an AI tool will make it into routine use.

To tackle these issues, we present ClinNetDx, a hybrid AI that integrates structured and unstructured input in one deep learning pipeline. Our model is distinct from previous approaches that address each input modality separately and connect tabular and image data for joint decision making. One of the main contributions of our approach is to convert structured real-valued input into two-dimensional arrays, which allows reformulation of the table as images. This allows the adoption of convolutional layers, originally developed for image processing, to detect spatial feature interactions between hematologic parameters. These feature maps are followed by early fusion with image-based embeddings produced by a parallel CNN streaming, which deals with the blood smear images. The concatenated form is subsequently input into a dense classification head. This design enhances prediction performance and facilitates the model's generalization across modalities and data sources. Empirical evidence indicates that integrating ocular morphology and laboratory data can yield more precise cancer diagnoses than either method alone [24].

Highly exact simulations act like a "black box" and are not ideal for medical purposes. Interpretability: the basis of confidence in clinical decision-support systems. To overcome this problem, ClinNetDx has created a dual-faceted explainability system with SHapley Additive exPlanations (SHAP) for structured data and Gradient-weighted Class Activation Mapping (Grad-CAM) for visual data. SHAP uses a game-theoretic attribution method to calculate attribute significance scores. Clinicians may use these scores to evaluate which laboratory

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indicators, such as aberrant platelet counts or increased white blood cell counts, have the greatest effect on prognosis. Grad-CAM produces a heatmap overlay on the original smear image that highlights the cell regions that influenced the output prediction, somewhat improving upon the prior approach when applied to image inputs. This transparency helps clinicians trust the system, making it easier to audit errors and comply with regulations. When combined, these techniques offer both local and global interpretability, which is important for systems that may one day operate in high-stakes environments such as cancer clinics [25].

Usability is also a major concern, since many promising AI models never make the leap from the research laboratory to the clinical arena, despite offering significant potential to improve patient care. This is typically due to bad interface design and a lack of harmony with healthcare workflows. To solve this problem, we have developed a Django-based web application, ClinNetDx, that enables real-time inference. A physician or lab worker may easily review a CBC report or a smear image to obtain a diagnosis or prognosis. It also allows the user to visualize Grad-CAM heatmaps or SHAP plots alongside the predictions, thereby improving interpretability. This implementation is readily scalable to hospitals, labs, and field clinics because it can be viewed and operated from a browser and does not require local GPU compute. Additionally, because the system is built to support REST API endpoints, it can be integrated with cloud-based analytics, mobile point-of-care diagnostic systems, or EHR systems [26].

ClinNetDx was evaluated using structured, image-based, publicly available, medically vetted hematologic diagnostic data. Each of the thousands of records in the collection includes both actual and fake CBC recordings. The smear images of several types of white blood cells were labelled in the image dataset. For both modalities, a typical pre-processing pipeline comprising z-score normalization, SMOTE for class balance, and imputation by standard for missing values in the tabular data was employed. Pixel normalization, augmentation, and scaling were all part of the image pre-processing. The optimal hybrid model was trained and tested using k-fold cross-validation to maximize generalizability. We discussed both performance and reliability-related measures, such as accuracy, precision, recall, F1-score, ROC-AUC, and, lastly, MCC. The spatial distribution of predictions across the known classes was examined using a confusion matrix, and visual validation of the model's reasoning process was enabled via interpretation tools. ClinNetDx's performance was consistent across folds and input types, suggesting that it might be used in clinical settings for the early detection and triage of ALL [27]. The following are the main contributions of this suggested study:

- A novel explainable multi-modal diagnostic framework, ClinNetDx, is proposed to jointly analyze structured complete blood count (CBC) parameters and peripheral ALL smear images within a unified hybrid CNN–DNN architecture, thereby enabling comprehensive prediction of Acute Lymphoblastic Leukaemia by exploiting complementary clinical information from heterogeneous data sources.

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- A novel tabular-to-spatial transformation mechanism is introduced that converts structured numerical blood test features into spatial representations suitable for convolutional neural networks, enabling the model to capture complex inter-feature relationships that are difficult to learn with conventional machine learning or fully connected networks.

- An integrated explainable AI framework is developed by combining SHAP for structured clinical data and Grad-CAM for ALL smear images, providing both feature-level and image-level explanations that enhance model transparency, improve clinician trust, and support evidence-based clinical decision-making.

- A robust multi-modal learning pipeline is designed by incorporating comprehensive data pre-processing, including normalization, missing-value imputation, and SMOTE-based class balancing, thereby improving the generalization capability and reliability of the proposed framework when handling heterogeneous and imbalanced clinical datasets.

- Extensive experimental validation demonstrates the effectiveness of the proposed framework using publicly available multi-modal hematological datasets under five-fold cross-validation. The proposed ClinNetDx framework consistently outperforms conventional machine learning and state-of-the-art deep learning models while maintaining high interpretability and robust predictive performance.

- The proposed framework is designed for practical clinical deployment, featuring a modular architecture that supports future integration with clinical decision-support systems, hospital information systems, and cloud-based healthcare platforms, thereby facilitating the translation of explainable AI into real-world ALL screening applications.

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The rest of this paper is structured as follows. Section 2 - Related Work: On AI-based medical diagnostics in haematological cancers, our approximations of AI in medical diagnostics were substantially challenged and comparatively discussed in the following sections. The proposed methodology, comprising data transformation, model architecture, pre-processing, and interpretability, is discussed in Section 3. Section 4 describes experimental settings, datasets, performance measures and the results discussion. Section 5 deals with implementation and deployments, and ethical considerations. Section 6 ends the paper with a discussion of limitations and directions for future work.

2. Literature Review

Ribeiro et al. [28] introduced LIME (model-agnostic local explanation method) to explain black-box classifiers. They demonstrate that regional surrogates can be used for the approximation of

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non-linear decision boundaries with sampling about these boundaries. LIME produces readable rules for humans that explain single predictions. This approach is particularly well-suited to clinical implementation, where transparency is essential. They showed that model trust increased for healthcare users when LIME explanations were provided. Their approach was not scalable, but it was the precursor to post-hoc explanation of AI systems in sensitive domains, including medicine.

Isa and Fernando [29] presented SHAP, a game theory-based unified explanation framework for model prediction explanation. Their method guarantees local accuracy and consistency, providing global and local explainability of model output. The SHAP approach leverages the Shapley values to allocate the feature importance equitably across all the inputs. In the healthcare sector, this has made it possible for clinicians to see how input features, here WBC or platelet counts, affect output. Due to its theoretical solidness and visual clarity, their framework has been widely utilized in clinical AI. Islam et al. [30] proposed a CNN-based model for blood cell classification based on microscopic images. They trained their model on public datasets, including BCCD, and reported high accuracy in distinguishing various types of white blood cells. The CNN method allowed for automatic feature selection, superseding manual classifications used in labs. However, their method did not have multi-modal input or interpretability. Their findings justify the application of cytology DL and motivated enhancement to hybrid diagnostic models.

Morano et al. [31] have shown that multi-modal deep learning outperforms disease detection by fusing image, signal and metadata streams. Their fusion network consisted of parallel CNNs for multimodality fusion, and output merging was performed on a decision layer to improve robustness. When put to the test with Alzheimer's data, the model's accuracy beat all single-input systems. According to those papers, clinical decisions tend to rely on more than one type of data. This demonstrates that combining blood test data with smear image benefits performance. Alom et al. [32] developed an ensemble of deep learning models for early cancer detection, with the use of several base learners, and adopted SHAP for interpretability. Their approach combined gradient boosting and neural networks, and they used SHAP values to understand how features contributed to the outputs. It was evaluated on EHRs, and the overall sensitivity and specificity were higher for the ensemble. This strategy reveals that ensemble learning could simultaneously achieve good performance and explainability in clinical AI.

Qoku et al. [33] presented a multi-modal biomedical prediction framework integrated with deep neural networks to amalgamate disparate clinical inputs. Their model divided structured tabular and unstructured textual data and sent them to different encoders, and then concatenated them in a late-fusion manner. The fusion was evaluated on biomedical data and consistently improved compared to the single-input networks. They demonstrate that the learned stable representations in both modalities are complementary to each other. Zhang et al. [34] performed a systematic review of machine learning and AI for cancer diagnosis in imaging, histology, and lab data.

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Their work classified multiple types of ML techniques and identified that CNNs and tree-based models were popular in histopathology and blood-based detection. They stressed the increasing need for interpretable systems in the clinic. Their results further motivate the development of hybrid models that can provide both accurate classification and explanations at the feature level. Choudhary et al. [35] proposed a deep feature fusion model for cell differentiation that integrated the results of four different CNNs to provide a combined diagnosis score. The model achieved better performances than classical classifiers and single CNNs on every release of leukocyte images. Their approach retained the feature richness of low- and high-level layers, and they showed that information residing at various depths in the CNN makes unique contributions to classification.

Qi et al. [36] investigated the use of SHAP in health care models and for both classification and regression tasks. They tested SHAP for patient mortality prediction, cancer detection, and a cardio-vascular risk model implemented by XGBoost and LightGBM. Their study found that SHAP explanations aid in detecting clinically relevant features and capture overfitting. They also presented SHAP for visual check-in dashboards. This provides validation for the use of SHAP on ClinNetDx for blood feature importance and embedding such explanations into the clinical interface for transparency. Almbaidin et al. [37] verified clinical AI for haematology, using digital image analysis and diagnostic annotations from practising haematologists. They examined the pragmatic use and technical efficacy of these systems. The accuracy, usefulness, and integration of the AI system into regular healthcare workflows were assessed. Pathologists prioritized explainability and visual clarity in their results. While ensemble learning approaches improve robustness and interpretative quality, they are seldom crafted for concurrent interpretability in cytological contexts and can elevate architectural complexity [38]. Reviews of cancer detection methods highlight the predominance of CNNs and tree-based approaches, but also emphasize the ongoing dearth of feature-level explanations for clinical use [39]. Although they lack unified post hoc interpretability and tabular biomarkers, deep feature fusion models show that multi-depth CNN representations improve discrimination [40].

Tülü and Kaya [46] developed the CONV-FAN-POX, which integrates the Fourier Analysis Network (FAN) with a deep learning architecture for monkeypox classification. Explainable AI was incorporated using Grad-CAM, which provides visual interpretations showing that the FAN layer successfully emphasizes lesion-specific features while suppressing irrelevant background information. This model achieves superior performance with fewer trainable parameters. Kaya et al. [47] proposed Fusion-Brain-Net, a hybrid transfer learning framework for automatic brain tumour classification that integrates three pre-trained deep learning models to improve feature extraction and reduce overfitting. Gursel and Kaya [48] proposed a deep learning framework for breast cancer classification using mammographic images. The model combined ConvNeXt for low-level feature extraction with a randomly initialized naïve Inception module to capture high-level representations, enabling accurate classification of breast regions of interest. Cheng et al. [49] developed a machine-learning-based predictive model for the early diagnosis of childhood

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leukaemia by analyzing nutritional biomarkers, leukaemia-related indicators, and granulocyte profiles from blood samples. Chen et al. [50] investigated the prognostic significance of dynamic carcinoembryonic antigen (CEA) trajectories in 578 patients with gastric cancer following curative resection. A k-means clustering algorithm was employed to classify perioperative CEA patterns into high-, medium-, and low-trajectory groups.

The primary research gaps identified from the literature review are: (i) Existing machine learning and deep learning methods have demonstrated promising performance in ALL diagnosis, most rely on either structured clinical data or ALL smear images independently, limiting their ability to capture complementary diagnostic information. (ii) Clinicians find it challenging to trust these models' predictions due to their lack of interpretability, and issues, including class imbalance, diverse data, and restricted clinical deployment, are still not well addressed. These limitations highlight the need for an explicable multi-modal diagnostic framework that amalgamates visual and organized medical data into a unified, dependable, and therapeutically beneficial methodology.

The assessed study demonstrates substantial advancements in explainable and multi-modal clinical AI, while also highlighting notable deficiencies that prompted the development of ClinNetDx. While LIME enhances user trust and provides explicit local explanations, its surrogate-based methodology is computationally intensive and has challenges in scaling to extensive clinical datasets. While SHAP offers theoretically grounded global and local interpretability, it is resource-intensive and encounters constraints when handling high-dimensional, multi-modal inputs because of its dependence on Shapley value estimations. Despite the impressive visual accuracy of CNN-powered models for blood cell categorization, their applicability in real-world diagnostic procedures is constrained due to their exclusive reliance on image features, coupled with a deficiency in interpretability and multi-modal integration. The bulk of prior research integrates just photos and metadata, or structured and textual inputs, rather than the clinically significant combination of smear images with CBC parameters. Nonetheless, multi-modal fusion frameworks routinely surpass single-input systems, as seen in research on Alzheimer's and general biological predictions. The ClinNetDx system seeks to address these deficiencies by including explainable AI, multi-modal feature learning, and a deployment-focused design, therefore allowing precise early prediction of acute lymphoblastic leukaemia.

3. Proposed Methodology

The detailed design and operation flow of the ClinNetDx system are given in this section. The approach is based on processing well-organized blood test data and medical images [51] within a hybrid deep learning architecture. Every part of the system, from data ingestion and transformation through feature extraction, interpretability, and interface deployment, is meant to be minimally invasive, designed to accommodate the clinician's speed and clinical relevance.

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The system architecture, data pre-processing pipelines, model formulation, and interpretability methods to support transparent and effective Acute Lymphoblastic Leukemia prediction are elaborated in the following subsections.

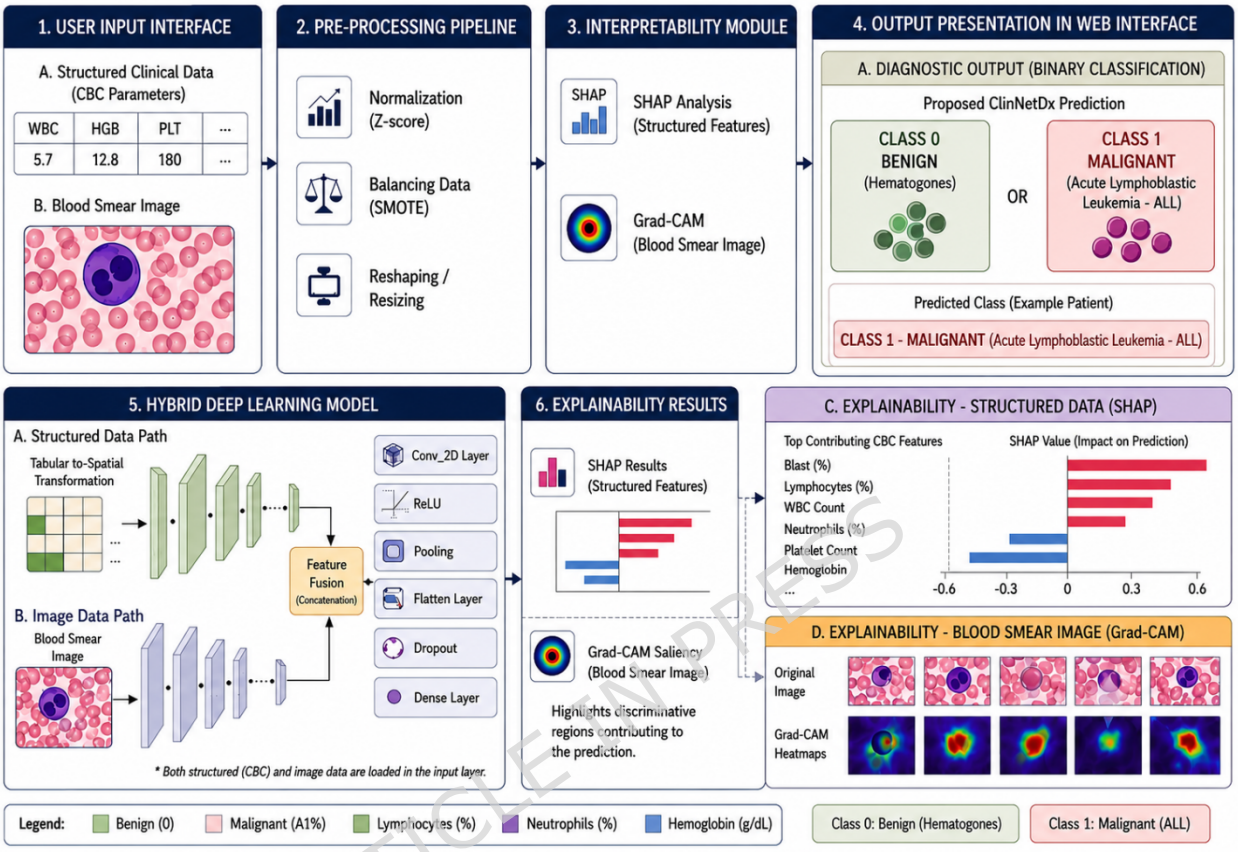


Figure 1: Overall workflow diagram of the proposed ClinNetDx system architecture.

3.1. Proposed ClinNetDx Architecture Overview

During the implementation, the ClinNetDx was developed as a blood-based modular and end-to-end diagnostic system for real-time Acute Lymphoblastic Leukemia prediction based on structured blood test data and stained ALL smear images. The proposed architecture, as depicted in Figure 1, consists of five modules: user input interface, pre-processing pipeline, hybrid deep learning model, plotting and compatibility layer, and output presentation module. The system starts with data collection from a Django web interface, where the user can upload either blood test values in a structured form or a high-resolution blood smear picture. These are responded to by parallel processing branches. For structured data, we normalize the data, balance and reshape the data into a two-dimensional pseudo-image matrix in a way that CNN processing can handle, while for image data, we normalize and resize it. Pre-processed inputs of both types (word and sentence) are sent through their own CNN stacks, where they are treated with convolutional, ReLU and pooling layers to extract spatial features. The resulting features are then flattened and

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combined into a dense decision layer of fully connected neurons with dropout regularization. Joint representation allows the model to learn cross-modality relations before making a binary prediction of whether ALL is probable. SHAP (for structured inputs) and Grad-CAM (for visual overlays on smear images) are integrated in the system for interpretability, to provide the clinicians with interpretations of how each input affected the prediction. The last output contains the prediction, confidence score, and an explanation visual, subsection rendered in the same web interface.

The base architecture of ClinNetDx is shown in Figure 1, where the input begins from the user interface to the framework and ends at the output screen. The modular architecture of such a system allows easy integration in the future with mobile health platforms, hospital health information systems (HISs) or RESTful APIs without changing the core prediction logic. This enables the utilization of ClinNetDx in many real-world clinical scenarios. The last stage of ClinNetDx involves binary diagnostic classification. For each patient sample, the conceptual framework predicts one of two clinically relevant categories: Benign (Hematogones) or Malignant (ALL). The methodology employs Grad-CAM for blood smear images and SHAP for structured clinical parameters to provide additional explanations beyond the predicted class. These outputs aim to provide meaningful, clinically interpretable decision-making rather than just providing a numerical prediction.

The proposed ClinNetDx framework was developed to address the complementary aspects of clinical information used in ALL diagnosis, rather than introducing additional complexity to the model. Clinicians never rely exclusively on a single source of information for diagnostic decisions in standard clinical practice; rather, they evaluate quantitative haematological indicators alongside morphological features observed in peripheral ALL smear images. Systems trained only on a single modality may overlook critical diagnostic information available in the other modality. Because of this, ClinNetDx uses a multi-modal architecture that simultaneously learns from microscopic images and structured laboratory measurements, enabling the extraction of complementary information to support more accurate diagnostic judgments. Therefore, rather than modeling an overly complicated model, the hybrid CNN–DNN architecture represents the varied properties of the clinical data.

3.2. Data Collection and Pre-processing Pipeline

The accuracy and consistency of the input data have a significant impact on a diagnostic model's prediction ability. ClinNetDx uses specific pre-processing pipelines for both structured numerical data and unstructured medical imaging data to achieve clinical-level reliability. White blood cell count (WBC), haemoglobin (HGB), platelets (PLT), and red cell indices are among the essential properties verified by the structured data flow after input validation and type checks. Depending on the feature distribution, the mean or median is

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used for imputation when there are missing values (see Equation (1)) in clinical datasets, where this is typically an issue. After verification, the data is further normalized by z-score transformation (see Equation (2)) to eliminate the scale effect among the channels and improve the stability of gradients during training. The system also tackles class imbalance that often arises when cancer-positive cases are not well represented, featuring SMOTE (see Equation (3)). This data augmentation via artificial oversampling generates new instances of the minority class through interpolation of existing instances, improving generalization of the model without overfitting to the dominant class.

$$x_i = \begin{cases} x_i, & x_i \neq \emptyset \\ \bar{x}, & x_i = \emptyset \end{cases} \quad (1)$$

Where, x_i denotes the clinical feature value, $\emptyset$ indicates a missing observation, and $\bar{x}$ represents represents the mean value used for imputation.

$$x_i' = x_i - \mu_i / \sigma_i \quad (2)$$

Where, x_i' is the normalized clinical feature, x_i is the original feature value, μ_i denotes the mean of the ith feature, and σ_i represents the standard deviation of the ith feature.

$$x_{new} = x_i + \lambda(x_j - x_i), \qquad \lambda \in [0,1] \quad (3)$$

Where, x_{new} denotes the newly generated synthetic sample, x_i is the selected minority-class sample, x_j represents one of the nearest minority-class neighbours, and λ is a randomly generated interpolation coefficient between 0 and 1.

In the case of image data, pre-processing requires resizing of input smear images to a uniform size suitable for the input layer of the CNN. The pixel values are also normalized between 0 and 1 to normalize the data in terms of brightness and contrast. Data augmentation, including flipping, rotation, and zooming, is performed to mimic real-world imaging variation and reduce overfitting during model training. All this previous work is executed automatically every time the user submits input through the interface and does not have to be set manually. The structured and image data are processed in parallel and are prepped to be fed to the hybrid model. This multi-mode pre-process pipeline enables asynchronous and multi-modal inference and reflects the flexibility that is adapted to a variety of clinical settings.

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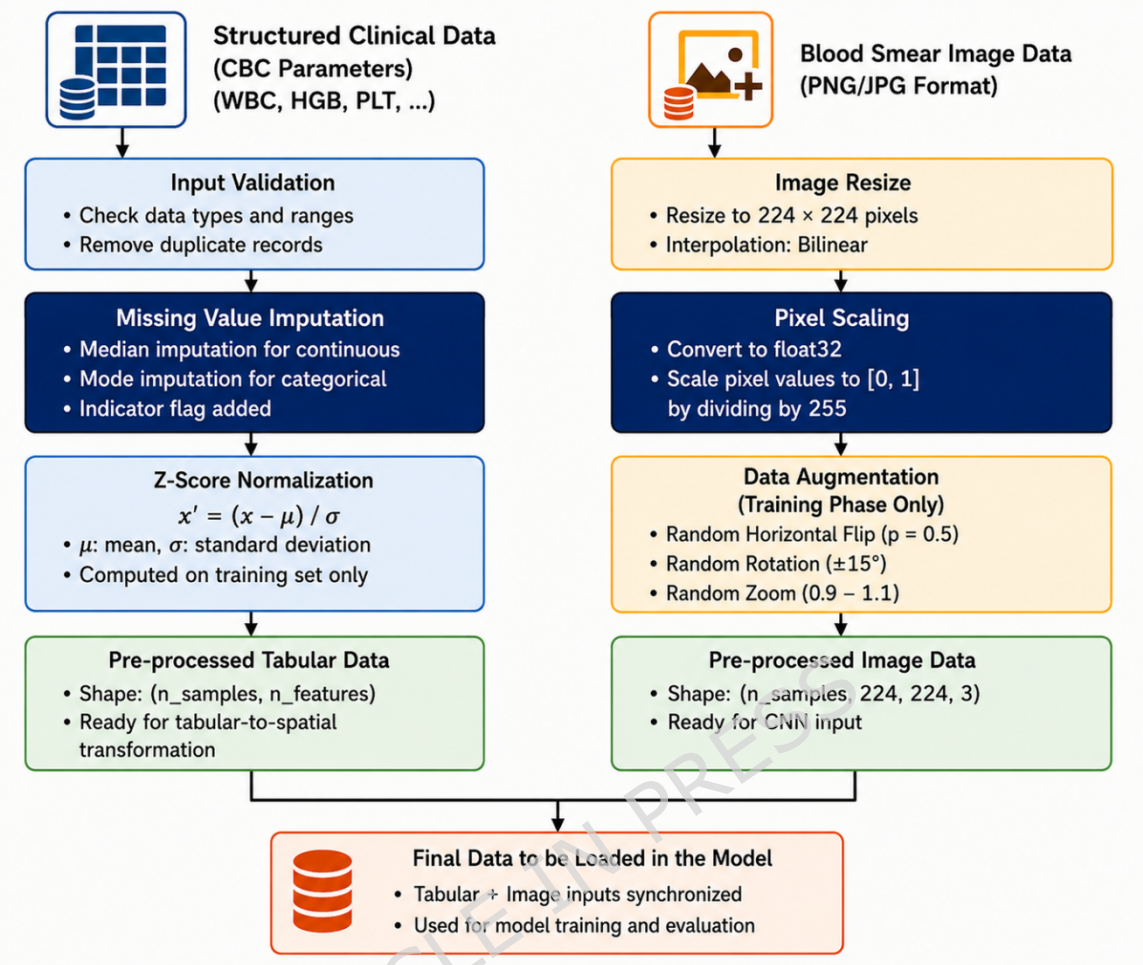


Figure 2: Data pre-processing spatial transformation pipeline for tabular and image inputs.

Figure 2 shows the complete data pre-processing pipeline in the context of ClinNetDx, including the standardization, augmentation, and class balance steps. Missing values were first detected by statistical inspection and then addressed by mean and mode imputation according to feature type. This assured data completeness before the application of normalization. Furthermore, following imputation, SMOTE was used to balance the minority classes without increasing the missingness. In the methodology section, further explanations and references were provided to make clear how the missing data were handled and how their influence on a model's performance was reduced. Overall, these techniques help ensure that the prediction engine is fed with genuine, clean, and representative data, leading to improvements in the diagnostic performance and real-world applications of the models.

Although numerous deep learning experiments have used the direct input of raw ALL smear pictures into networks, medical images collected under varied laboratory circumstances are usually variable in terms of lighting, staining quality, image resolution, magnification and background noise. Such discrepancies might weaken the stability of the features and

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consequently impact model generalization. Therefore, standard pre-processing operations were incorporated to produce consistent input representations and improve robustness against acquisition variability. These operations do not alter the underlying diagnostic characteristics of the images; rather, they standardize image quality and enable the model to learn more reliable and clinically meaningful representations.

3.3. Tabular-to-Spatial Data Transformation

One of the main technological contributions of ClinNetDx is the ability to make CNNs applicable to structured blood test data by translating it to a spatial form. Usually, the tabular data is handled with standard machine learning techniques or dense neural networks, where individual characteristics are whitened independently. However, such models may fail to capture inter-feature connections, which are important for complicated diagnostic patterns. For this purpose, we designed a tabular-to-spatial translation module, which transforms a one-dimensional clinical feature vector into a two-dimensional matrix.

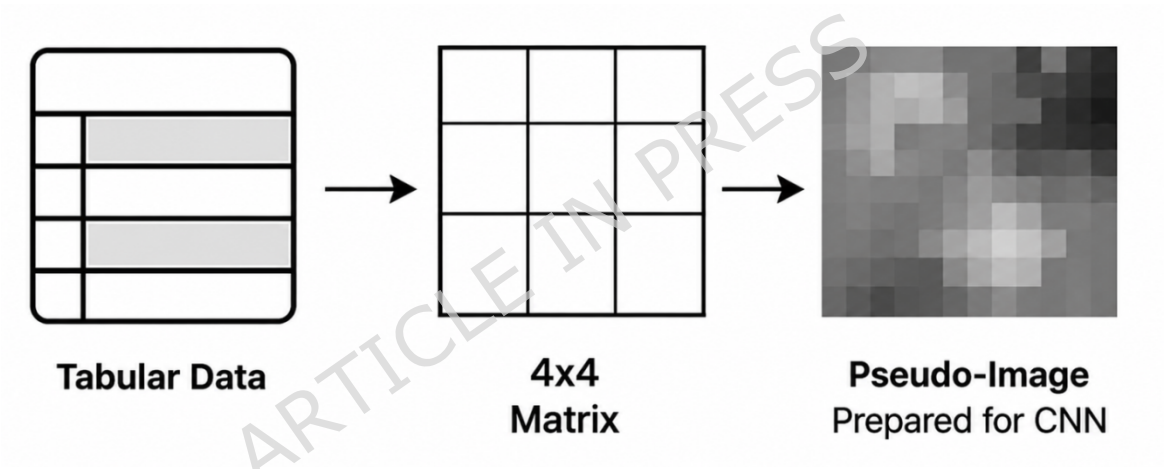


Figure 3: Tabular-to-spatial transformation for structured clinical data.

Figure 3 illustrates how structured numerical input is transformed into a matrix and is ready for convolutional processing. For instance, when we have 16 features, the vector is reshaped into a 4×4 matrix, while keeping the order of features and placing the related hyperparameters close together. This pseudo-panorama is then sent through convolutions with local filters that can detect hierarchical and spatially dependent patterns among local feature clusters. This approach capitalizes on the ability of the CNN to learn feature locality, for example, combining abnormalities in white blood cell count and haemoglobin levels, which would be considered separate entities. For the sake of coherence, feature layout in the matrix is preserved for all the samples of the same class, providing the model with consistent spatial relationships. This transformation is done in real time and is hidden from the user who enters structured data in the web interface, with no knowledge about how this reshaping is done internally. When converted, the pseudo-image flows through a network stack with the

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same architecture as the CNN for real-image data, which provides unified feature extraction across modalities.

$$I = R(x) = \left[x_{4(i-1)+j}\right]_{i,j} \in \mathbb{R}^{4\times 4} \tag{4}$$

This compactly expresses that the pseudo-image I is obtained by a reshaping operator $\mathbb{R}$ acting on the tabular vector x as given in Equation (4). This strategy significantly broadens the applicability of CNNs to more general structured data beyond classic image-type domains and improves our ability to interpret predictions given for structured data when combined with SHAP-based explanations. This leads to a more comprehensive depiction of clinical symptoms, enabling the model to uncover intricate diagnostic signals that conventional tabular models may overlook.

The structured CBC measures are numerical variables that have significant correlations, but no intrinsic spatial organization. These variables are processed individually by conventional fully connected networks and might not be able to adequately capture local feature interactions. The proposed tabular-to-spatial transformation aggregates correlated clinical variables into a fixed two-dimensional representation, which allows convolutional filters to learn local dependencies among haematological parameters while maintaining a unified learning strategy over structured and image modalities. This transformation enhances the feature representation without separate learning frameworks for distinct data kinds.

3.4. Hybrid CNN-DNN Model Design

ClinNetDx is based on a hybrid deep learning architecture combining CNN and DNN for handling structured clinical data and stained ALL smear pictures. The architecture is motivated by the need to retain modality-specific cues at the early levels while allowing for joint reasoning at the time of classification. The system aims to bridge the structural divide between numerical inputs and images by performing a tabular-to-spatial transformation, which converts structural data into 2D matrices of pseudo-images, and represents tabular features with the help of spatial filters by CNNs. The modified input data is propagated through a dedicated CNN pipeline formed by convolution-based filters to extract patterns using Bi-GRU sequential encoding, ReLU activation functions to introduce non-linearity, and max-pooling blocks, which help to reduce dimensionality but retain significant information. Alongside, raw ALL smear images are fed into a second CNN stream, which is tailored for the extraction of morphological features such as nuclear irregular contours, high nucleus-to-cytoplasm ratios, and abnormal leukocyte clustering, characteristic features of hematologic cancer.

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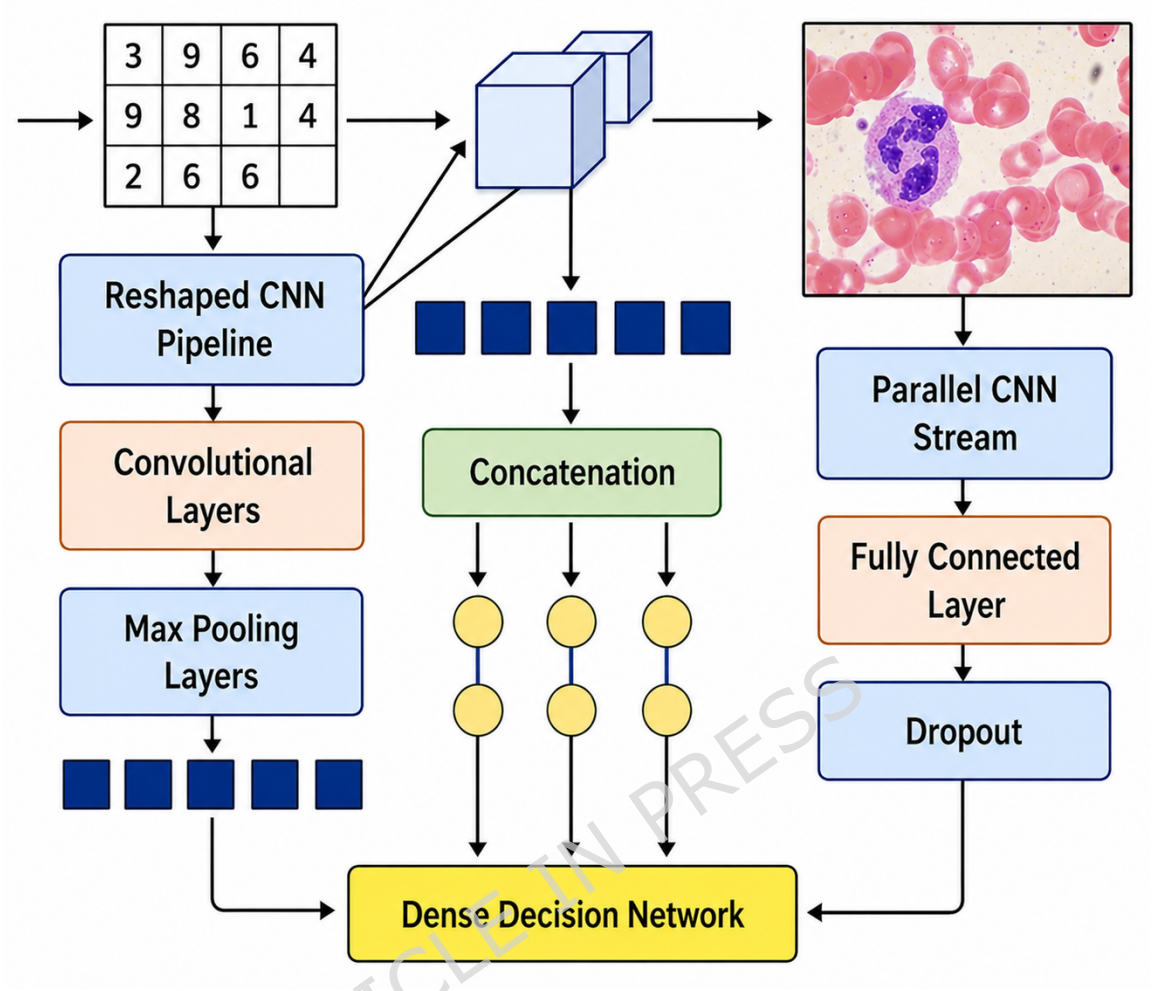


Figure 4: Proposed Hybrid CNN-DNN model architecture for multi-modal classification.

When both input streams have occupied their symmetric CNN paths, the feature maps are then transformed into vectors and appended into a single representation. Concatenate this vector with the third sentence representation and feed the combined vector into the DNN for classification, which is a set of fully connected layers with dropout to avoid overfitting. We use batch normalization for fast training and stable learning dynamics. The output layer is the sigmoid activation to be able to output a probability score for binary classification, i.e. ALL presents or not! The model is trained to minimize the binary cross-entropy loss on updating all the parameters with the offset and learning rate values in the optimization strategy of Adam, and early stopping is used to prevent the model from overtraining and to converge. This structure facilitates the model to learn modality-specific low-level features independently, yet fuse them at a later stage for high-level decisions.

Network Layer	Configuration	Output

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Input	Input size = 32 × 32 × 1 Kernel size = 3 × 3, Stride = 1,	32 × 32 × 1
Conv2D-1	-	32 × 32 × 32
MaxPooling2D-1	Output channels = 32, Activation = ReLU Pool size = 2 × 2, Stride = 2 Kernel size = 3 × 3, Stride = 1,	16 × 16 × 32
Conv2D-2	-	16 × 16 × 64
MaxPooling2D-2	Output channels = 64, Activation = ReLU Pool size = 2 × 2, Stride = 2 Kernel size = 3 × 3, Stride = 1,	8 × 8 × 64
Conv2D-3	-	8 × 8 × 128
MaxPooling2D-3	Output channels = 128, Activation = ReLU Pool size = 2 × 2, Stride = 2	4 × 4 × 128
Flatten	Converts 3D feature map to 1D vector	2048
Dense-1	Units = 256, Activation = ReLU	256
Dropout-1	Dropout rate = 0.5	256
Dense-2 (Output)	Units = 1, Activation = Sigmoid	1

Table 1: Feature extraction configuration of the CNN used in ClinNetDx Network Layer Configuration Output Feature Map.

The full configuration, including the convolutional layers, kernel sizes, activation functions and output dimensions of the CNN feature extractors, is presented in Table 1. Figure 4 shows the dual-stream architecture. The model combines structured and image data before performing classification. Notably, architecture also enables partial input inference, making ClinNetDx perform well even when only one of the modalities is available. This resilience is crucial for clinical implementation when imaging or lab tests may be inaccessible or delayed, thereby ensuring adaptability and practical feasibility across diverse clinical situations.

3.5. Explainable AI Integration

Interpretability is a key component of clinical AI, as acceptance and decision-making are strongly based on trust and transparency. This is addressed by ClinNetDx using SHapley Additive exPlanations (SHAP), an advanced explainable AI (XAI) approach for structured blood test data. SHAP assigns a “contribution score” for each patient-feature pair to the final forecast. It allows doctors to identify which laboratory parameters, such as white blood cell count, haemoglobin levels or platelet counts, were most relevant in the diagnosis of ALL. The connection is shown in color and signed to indicate whether characteristics are positive or negative in a horizontal bar plot of SHAP values. A high white cell count and a low

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haemoglobin would likely signal a cancer-positive result, and a normal platelet count would counteract this. These simple explanations provide real-time interpretability at the point of treatment, allowing physicians to assess if the model's actions are consistent with medical expectations. Importantly, the SHAP result is shown immediately in the web-based application, alongside clinical forecasts, so physicians do not need to use other tools or systems. This incorporation enables open, informed decision-making and encourages wider adoption of AI-based diagnostics. Figure 5 illustrates an example SHAP output for the clinical case, highlighting the effect of structured input characteristics on the final model conclusion. ClinNetDx injects numerical transparency into its basic functions, bringing clarity to the diagnostic process by making it intelligible, dependable, and clinically actionable.

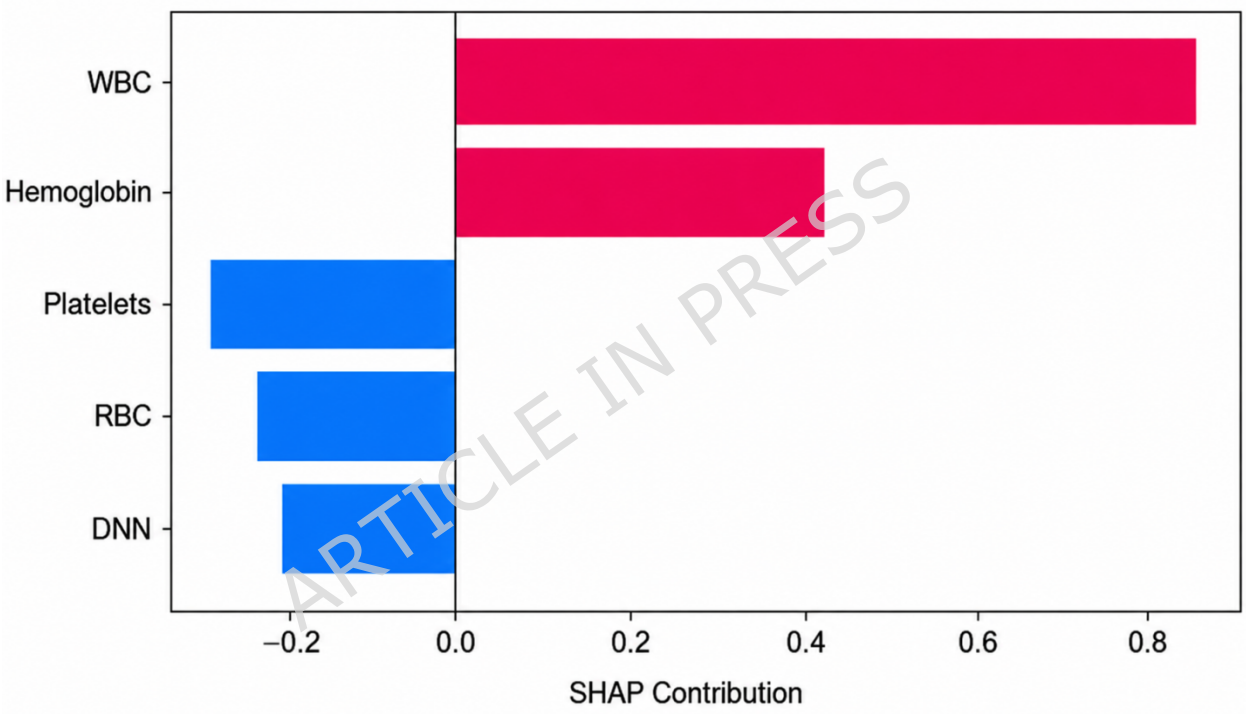


Figure 5: SHAP output for structured data explainability in ClinNetDx.

The proposed Boosted-PSO algorithm, as given in Algorithm 1, has the greatest impact on the accurate classification results. Here, X_{img} denotes the microscopic ALL smear images, X_{tab} represents the tabular CBC features, and Y is the ground-truth class label. $\hat{Y}$ is the predicted class, P is the confidence score, and E refers to the combined explainability maps.

Algorithm 1 Proposed ClinNetDx ALL Classification Algorithm

Require: Microscopic ALL smear images X_{img}, tabular CBC features X_{tab}, and ground-truth labels Y for ALL classification.

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Ensure: Predicted class $\hat{Y}$, confidence score P, and explainability maps E including SHAP values and Grad-CAM heatmaps.

Step 1: Pre-Processing

Tabular feature normalization: $f_i' = (f_i - \mu_i)/\sigma_i$ and Image resizing: $I' = resize(I; H, W)$

Step 2: CNN Feature Extraction

Convolution: $F_l = ReLU(W_l * I' + b_l)$ with the customized layer parameters

Pooling: $P_l = maxpool(F_l)$, Flattening: $Z_{img} = flatten(P_l)$ and dropout 0.2

Step 3: Bi-GRU Sequential Encoding

Forward: $h_t^{\rightarrow} = GRU(S_t, h_{t-1}^{\rightarrow})$

Backward: $h_t^{\leftarrow} = GRU(S_t, h_{t+1}^{\leftarrow})$

The resulted combined state is given as $H_t = [h_t^{\rightarrow}; h_t^{\leftarrow}]$

Final sequence embedding: $Z_{seq} = H_t$

Step 4: Feature Fusion & Classification

Fusion: $Z_f = Concat(Z_{seq}, X_{tab})$ with the class logits

$O = W_c(Z_f) + b_c$ using SoftMax with $\hat{Y} = argmax(P)$ prediction argument

Step 5: Gradient-Guided ACO Optimizations

Probability of choosing update $P(\Delta\theta_i) = \left(\tau_i^{\alpha} \cdot \eta_i^{\beta}\right) / \sum_j \left(\tau_j^{\alpha} \cdot \eta_j^{\beta}\right)$ and $\eta_i = |g_i|$ (Adam Gradient Magnitude) with parameter update

Pheromone update as $\theta_i \leftarrow \theta_i + \Delta\theta_i$ and $\tau_i \leftarrow (1-\rho)\tau_i + 1/(1+L)$ using cross entropy

Step 6: Explainability

SHAP values: $\emptyset_i = SHAP(f_i | model)$

Grad-CAM: $\alpha_k = (1/Z) \cdot \sum_x \sum_y (\partial score_c / \partial A_k(x,y))$

Step 7: Output Generation

Prediction = $\hat{Y}$ and Confidence = P and Explanation =$E = \{\emptyset_i, M(x,y)\}$

The f_i is the i[th] tabular feature with mean μ_i and standard deviation σ_i and f_i' is its normalized form. I' is the resized image, W_l and b_l are the convolution layer weights and biases, and F_l is the ReLU-activated feature map. P_l is the pooled feature map, Z_{img} is the flattened CNN feature vector, and S_t is the input state at time step t for the Bi-GRU. $h_t^{\rightarrow}$ and $h_t^{\leftarrow}$ denote the forward and backwards hidden states, and H_t is their concatenation. Z_{seq} is the final sequence embedding, Z_f is the fused feature vector, W_c and b_c represent the classifier parameters, and O is the output logits. τ_i is the pheromone value, η_i is the heuristic gradient magnitude, α and β are the ACO

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exponents, ρ is the pheromone evaporation rate, $\Delta\theta_i$ is the parameter update, θ_i is the model parameter, and L denotes the loss. $\emptyset_i$ are the SHAP feature-attribution values, A_k is the kth activation map, α_k is the Grad-CAM weight derived from the gradients, and $M(x,y)$ is the final heatmap used for visual explanation.

4. Results and Discussion

4.1. Experimental Setup

This section presents an in-depth investigation of the ClinNetDx system, discussing dataset characteristics, architecture, measures for performance, and comparative analysis with base models. The goals of this study were (1) to evaluate the diagnostic performance of the hybrid method that consists of the CNN for the structured data and the RCNN algorithm for the image-level data, and (2) to assess the interpretability, deployment readiness and clinical alignment of the system. We start with a description of the multi-modal dataset and an overview of the design and structure of the data, which consists of a combination of tabular blood test results and images of peripheral blood smears. The model architecture, training pipeline, and deployment environment are then described. Results–In the following, we describe the chosen evaluation metrics–accuracy, precision, recall, the ROC-AUC (Area under the Curve of the Receiver Operating Characteristic) of a model and the Matthews Correlation Coefficient (MCC)–and the results obtained from 5-fold validation. Interpretations are also provided for the system predictions with the help of SHAP and Grad-CAM for structured features and image-based explanations, respectively. Finally, we benchmark ClinNetDx against 13 state-of-the-art machine learning and deep learning methods under the same data and experimental settings. Results show that ClinNetDx is superior to regular classifiers and CNN-based models and that it can make transparent and clinically sound predictions. The integrated nature of this system places it as a feasible decision support tool to be deployed for early ALL screening.

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4.2. Data Split for Training and Validation

The proposed ClinNetDx is a system created from multi-modal data in the form of structured numerical features and unstructured visual data that mirrors clinical diagnosis processes. The structured dataset contains 5000 anonymized complete blood count (CBC) records with 9 diagnostic parameters such as white blood cell (WBC) count, haemoglobin levels, platelet count, and red cell indices [53], as given in Table 3. These features were extracted and formatted as those in the UCI Machine Learning Repository. Each observation was classified as cancer-positive or cancer-negative depending on clinically relevant thresholds and trends. Additionally, 2500 ALL smear images were obtained from public haematology datasets like Kaggle's BCCD, as well as open medical image repositories [51], Figure 6 shows the sample images ALL dataset. These images were pre-annotated by experts and stored as high-resolution images with ground truth information. To mimic the dual-modality diagnostics, a subset of 2000 matched (tabular,

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image) records was generated. Data was stratified into Train 80%, Valid 10%, Test 10%, with class balance within each fold.

Structured data pre-processing includes z-score normalization, missing value median imputation and Synthetic Minority Over-sampling Technique (SMOTE) for class imbalance [52]. For picture data, pre-processing included downsizing to 128×128 pixels, scaling the pixel values and augmentations such as random flips, rotations and zooms. This allowed to generalize across patient-dependent variability and imaging artifacts. All the detailed settings of the dataset splits, including the input dimensions, class ratios and modality combinations are reported in Table 2. This foundation allowed ClinNetDx to learn from statistical biomarkers and morphological indications, and hence to generate more informed and interpretable diagnostic outputs. As a final testing set, we prepared a dual-input synchronized dataset that represents the fully integrated real-world hematologic diagnostic, where both quantitative information and a visual examination are commonly performed. It was this exhaustive design of the dataset that allowed the hybrid architecture to significantly outperform the typical models that use only one input type.

Data Type	Source	Samples	Features / Resolution	Positive (%)	Use
Tabular (CBC Data)	UCI ML Repository-inspired [53]	5000	16 numerical features	37%	Training/Test
Peripheral Blood Smear Images	Acute Lymphoblastic Leukemia (ALL) Image Dataset (Kaggle) [51]	2500	128×128 RGB	42%	Training/Test
Combined –Multi-modal Dataset	Synchronized Dual Input Set [31]	2000	CBC + Blood Smear Images	40%	Final Testing

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Table 2: Summary of datasets used for structured and image input.

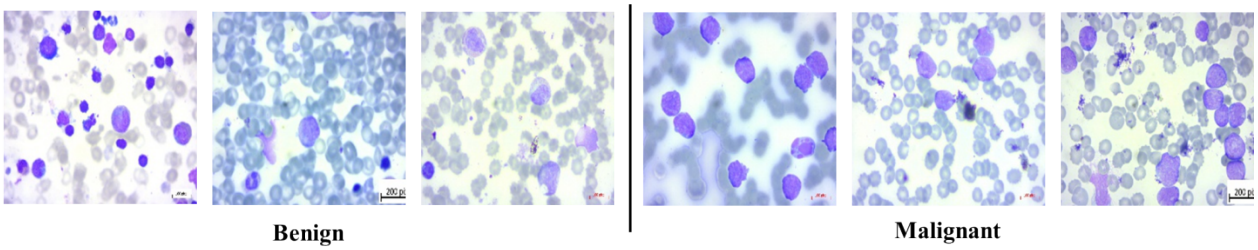


Figure 6: Sample images of the ALL image dataset.

Age	Gender	Hemoglobin	Platelet_ Count	White_ Blood_ Cells	Red_ Blood _Cells	MCV	MCH	MCHC
68	Female	10.4	180000	5700	3.7	77	25	32

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25	Male	13.8	320000	7500	5.4	92	30	32
57	Male	13.5	370000	8500	5.1	90	29	32`
….	….	….	….	….	….	….	….	….
….	….	….	….	….	….	….	….	….
55	Male	13.8	350000	8000	5.3	92	30	33

Table 3: Samples of CBC Dataset.

4.3. Implementation Details and Model Hyperparameters

Building and deploying ClinNetDx necessitated a single pipeline that supported end-to-end operations, including model training, optimization, interpretability embedding, and interface deployment. The whole system was developed using Python and TensorFlow to implement the model and Django as the backend framework for its web application. The model was trained on a high-end machine with an NVIDIA RTX 3080 GPU, 32 GB RAM, and an Intel i9 CPU, which facilitated concurrent training of multi-modal data streams. The tabular structure inputs were first converted into 32×32 grayscale pseudo-images to be compatible with the structure of the convolutional layers. ALL smear images were resized to 128×128×3 and used as visual input to a separate corresponding CNN pipeline. Following input of each modality was forwarded by 3 convolutional layers (in kernel size of 3×3), followed by a max pooling layer and ReLU activation function. The outputs of the two branches were flattened and concatenated and sent to a fully connected layer, followed by dropout regularization (rate 0.5) and sigmoid classification. Short-circuiting with patience 15 was utilized to avoid overfitting. All the training settings, including the optimizer selection, learning rate, epochs and dropout strategy, are presented in Table 4

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Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Cross-Entropy
Batch Size	32
Epochs (Patience)	Max 50 (Patience = 15)
Dropout Rate	0.5
Input (Tabular)	32 × 32 single-channel Input (Image)
Input (Image)	128 × 128 × 3
Deployment Format	TensorFlow Saved Model + API

Table 4: Hyperparameters and Training Setup.

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The overall model design in terms of layer organization, size, and number of filters in the convolutional blocks is described in Table 5. After training, the model was exported to TensorFlow's SavedModel format and imported to the Django web backend via REST APIs. The predictions were supplied asynchronously over AJAX to allow real-time feedback on the front end. Consequently, SHAP and Grad-CAM were integrated as the main explainability engines for structured and picture data, respectively. These were built directly into the Django interface. Figure 11 shows the implemented online interface. Clinicians may input tabular data or smear photos to obtain a binary forecast and visualize feature and visual attributions. The strong coupling between model inference and clinician-facing visualization moves ClinNetDx from an experimental prototype to a deployable clinical application, balancing performance and usability in diagnostic processes.

Network Layer	**Configuration**	**Output Feature Map Dimension**
Input	Input size = 32 × 32 × 1 Kernel size = 3 × 3, Stride = 1,	32 × 32 × 1
Conv2D-1	Output channels = 32, Filters = 32,	32 × 32 × 32
MaxPooling2D-1	Activation = ReLU Pool size = 2 × 2, Stride = 2 Kernel size = 3 × 3, Stride = 1,	16 × 16 × 32
Conv2D-2	Output channels = 64, Filters = 64,	16 × 16 × 64
MaxPooling2D-2	Activation = ReLU Pool size = 2 × 2, Stride = 2 Kernel size = 3 × 3, Stride = 1,	8 × 8 × 64
Conv2D-3	Output channels = 128, Filters = 128,	8 × 8 × 128
MaxPooling2D-3	Activation = ReLU Pool size = 2 × 2, Stride = 2	4 × 4 × 128
Flatten	-	2048
Dense-1	Units = 256, Activation = ReLU	256
Dropout-1	Dropout rate = 0.5	256
Output Layer	Units = 1, Activation = Sigmoid	1

Table 5: Model configuration with output feature map dimension.

4.4. Evaluation Metrics

We comprehensively evaluated the predictive performance and clinical value of ClinNetDx using seven key assessment measures, including accuracy, precision, recall (sensitivity),

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specificity, F1-score, ROC-AUC (area under the ROC curve) and MCC (Matthews correlation coefficient) [54]. Specific selections of performance metrics are based on the robustness in assessing balanced and imbalanced clinical data sets. For the test set, we applied a 5-fold cross-validation approach, with each fold experiencing equal pre-processing and model evaluation. ClinNetDx achieved a mean accuracy of 96.8, precision of 95.2, recall of 97.4, F1-score of 96.3, and ROC-AUC of 98.1, with low standard deviations across folds, demonstrating the stability and generalizability of the model.

The detailed results are shown in Table 6 Representation of the diagnostic consistency was also demonstrated by the confusion matrix averaged by the folds, where both the false positives and the false negatives were observed at relatively low frequencies.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{5}$$

$$Precision = \frac{TP}{TP+FP} \tag{6}$$

$$Recall = \frac{TP}{TP+FN} \tag{7}$$

$$F1-score = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \tag{8}$$

$$Specificity = \frac{TN}{TN+FP} \tag{9}$$

$$Sensitivity = \frac{TP}{TP+FN} \tag{10}$$

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Metric	Mean (%)	Std. Dev.
Accuracy	96.8	±1.1
Precision	95.2	±1.4
Recall	97.4	±0.9
Specificity	96.2	±1.3
F1-Score	96.3	±1.2
ROC-AUC	98.1	±0.6
MCC	94.7	±1.5

Table 6: Hyperparameters and training setup.

Where, TP denotes the number of correctly predicted positive samples, TN represents correctly predicted negative samples, FP refers to negative samples incorrectly predicted as positive,

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and FN denotes positive samples incorrectly predicted as negative. The results of the precision, recall, and F1-score listed in Table 7 are fold-wise distributions with all of them being over 95%, further proving the reliability of the model. Table 8 shows the ROC-AUC for single-modality and fused-input modalities configurations, and the hybrid structure of ClinNetDx yields the best performance.

Fold	Precision (%)	Recall (%)	F1-Score (%)
Fold 1	94.7	96.8	95.7
Fold 2	95.3	97.6	96.4
Fold 3	94.9	97.1	96.0
Fold 4	95.0	96.9	96.1
Fold 5	95.2	97.4	96.3

Table 7: Precision-Recall metrics per fold**.**

Input Type	ROC-AUC (%)	Precision (%)	Recall (%)	F1-Score (%)	MCC (%)
Tabular Only	96.1	93.4	95.1	94.2	91.3
Image Only	96.8	94.8	96.5	95.6	92.8
Proposed ClinNetDx (Both Tabular + image)	98.1	95.2	97.4	96.3	94.7

Table 8: ROC-AUC comparison across input modalities**.**

4.5. Quantitative and Qualitative Analysis

For feature importance interpretation on structured input data, SHAP explainability was added in addition to pure statistical analysis. Additionally, higher counts of WBC, lower values of hemoglobin and platelet were the three parameters with the most significant impact on our prediction results (see Figure 7). In addition, the overall diagnostic performance is visually represented in Figure 8, showing a heatmap for the confusion matrix, clearly corresponding to model quality for predicted vs true classes. These figures, along with quantitative

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performance, corroborate ClinNetDx's functioning as a high-accuracy classifier and a clinically interpretable DSS.

Model	Type	Accuracy (%)	F1-Score (%)	Explainability
Logistic Regression [37]	Tabular	89.7	88.5	No
Decision Tree [51]	Tabular	87.4	86.9	Partial
Random Forest [24]	Tabular	90.6	89.4	Partial
SVM [54]	Tabular	91.4	90.7	No
XGBoost [23]	Tabular	92.7	91.8	Yes (SHAP)
DNN [27]	Tabular	92.1	91.6	Partial
CNN (Simple) [24]	Image	90.5	89.3	Grad-CAM
ResNet18 [28]	Image	93.5	93.0	Grad-CAM
InceptionV3 [14]	Image	94.1	93.6	Grad-CAM
MobileNetV2 [15]	Image	91.8	91.0	Grad-CAM
VGG16 [26]	Image	92.5	91.9	Grad-CAM
Hybrid CNN-DNN (No XAI)	Tabular + Image	95.1	94.5	No
Proposed (ClinNetDx)	**Tabular + Image**	**96.8**	**96.3**	**SHAP + Grad-CAM**

Table 9: Performance comparison of proposed models with other popular models.

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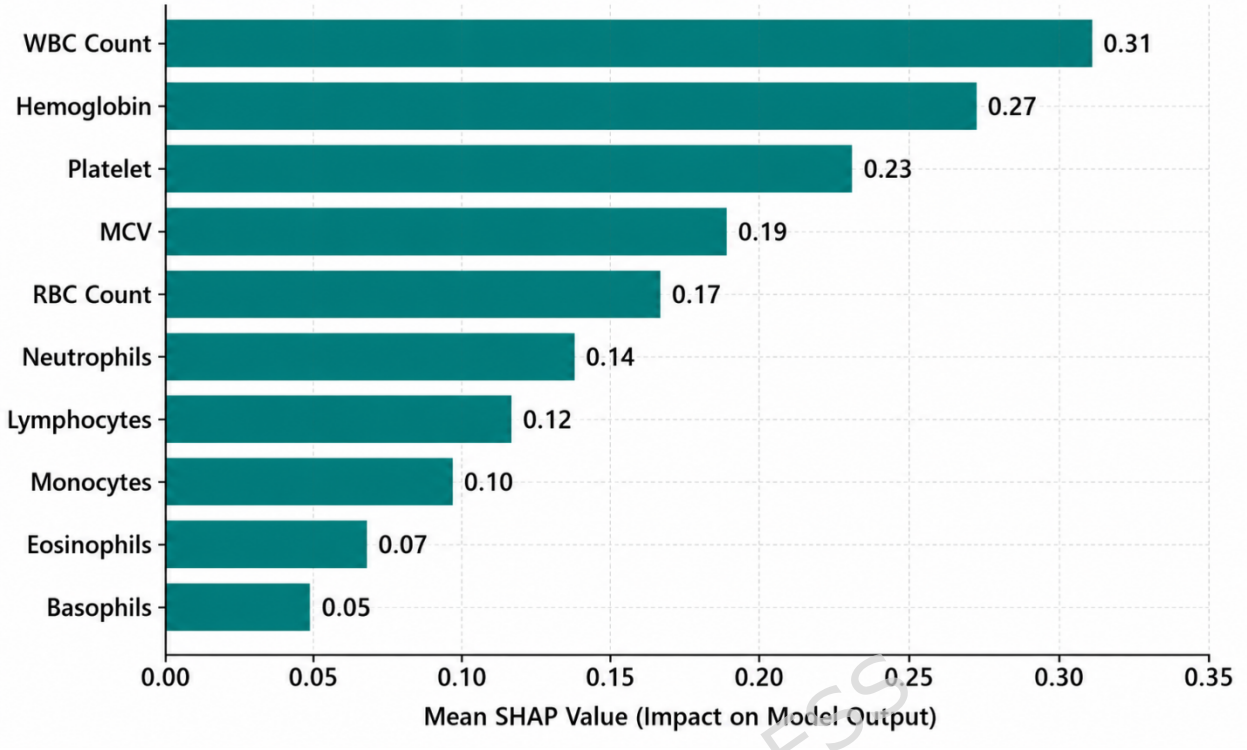


Figure 7: SHAP output visualizing feature contributions for structured input prediction.

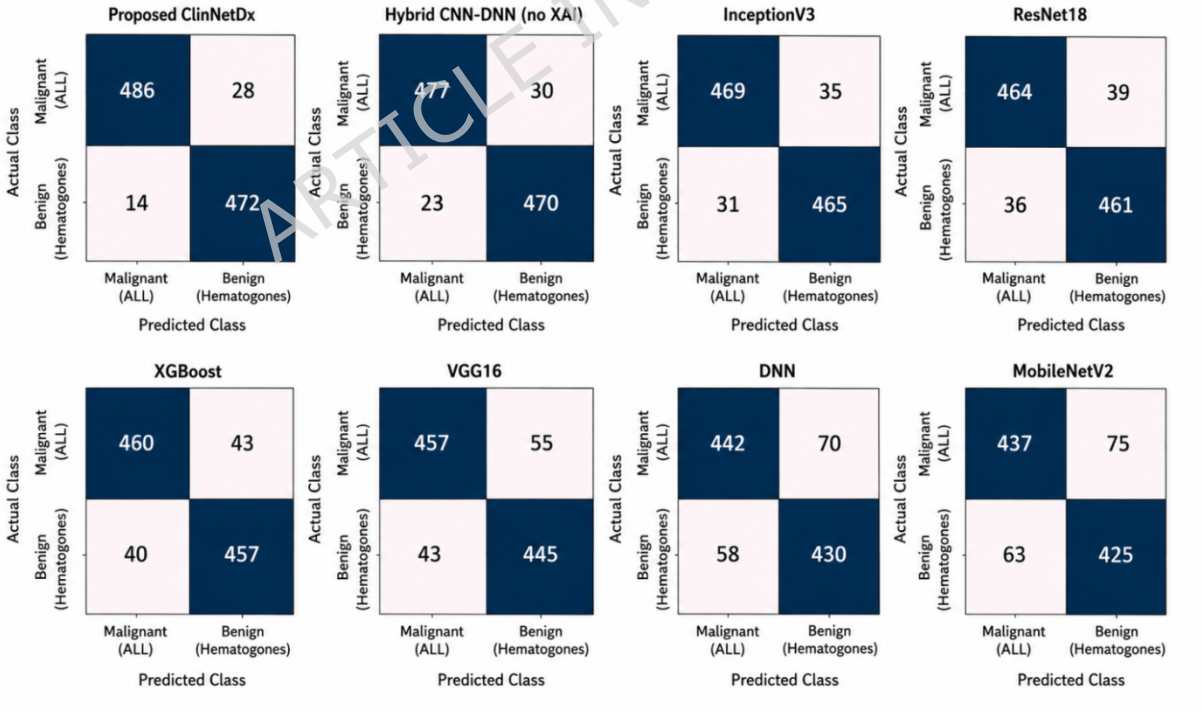


Figure 8: Confusion matrix heatmap for ClinNetDx averaged over 5-fold cross-validation.

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To compare the performance of ClinNetDx with other diagnostic models, we tested 13 alternative models trained from the same dataset splits and pre-processing conditions, as presented in Table 9. These baselines consist of classical classifiers like (linear) logistic regression, (linear) SVM, decision trees, random forests, and XGBoost for tabular data, as well as deep learning with fully connected DNNs. Convolutional architectures: ResNet18, VGG16, InceptionV3 and MobileNetV2 were tested for image-based comparisons. Furthermore, the two ablated sectors of the ClinNetDx model were considered: the one without SHAP and Grad-CAM explainability layers and the one that does not include the tabular-to-spatial transform module. Comparison of the performance results is visualized in Figure 9.

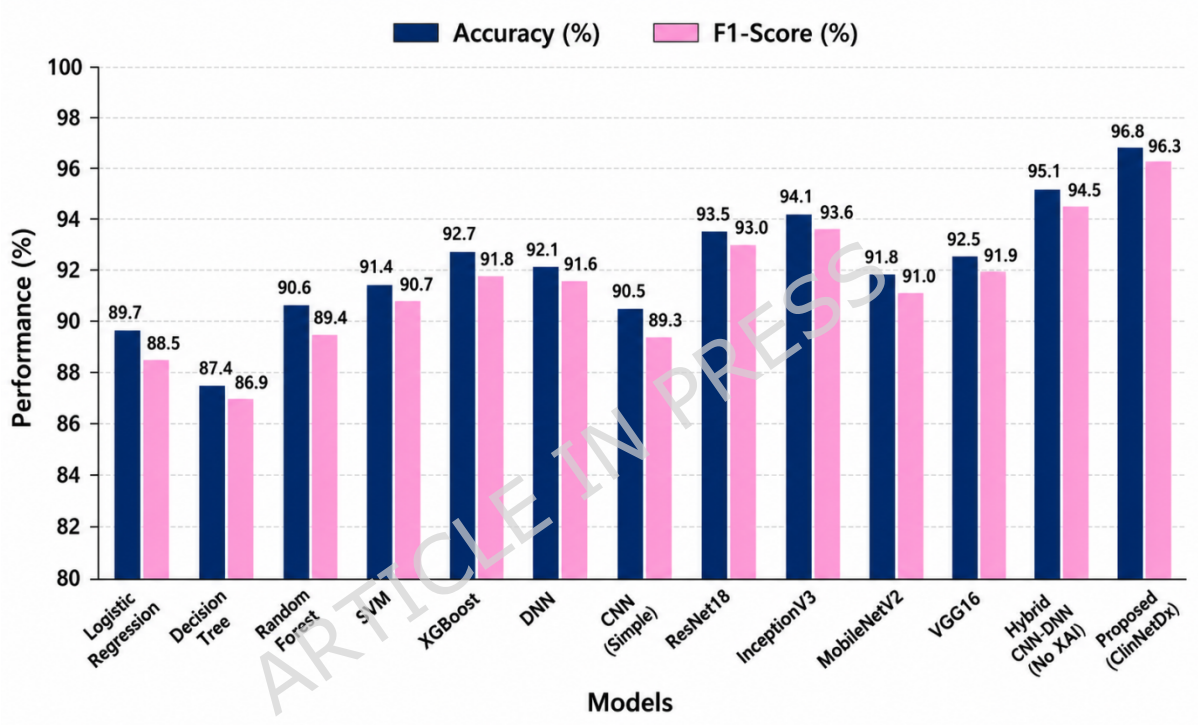


Figure 9: Performance comparison of proposed models with other popular models.

Vision Transformer architectures have shown encouraging performance in image classification tasks; their effectiveness is generally associated with large annotated datasets, high computational requirements, and image-centric learning frameworks. In contrast, the proposed ClinNetDx framework is specifically designed for multi-modal clinical diagnosis by integrating structured CBC parameters and peripheral blood smear images within a unified hybrid CNN–DNN architecture. The proposed tabular-to-spatial translation facilitates convolutional feature learning from structured clinical data. SHAP and Grad-CAM give complementary explanations and promote model transparency. Consequently, ClinNetDx offers a balanced framework that combines multi-modal learning, explainability, computational efficiency, and practical clinical applicability.

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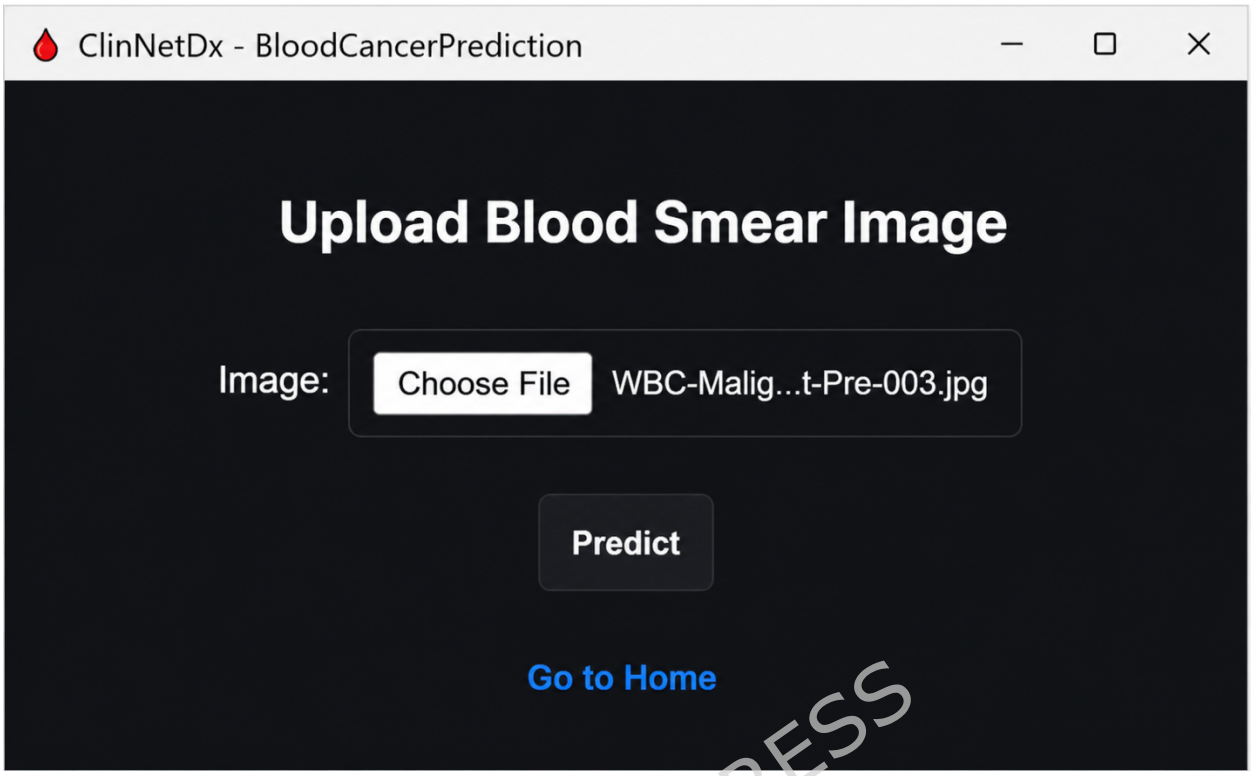
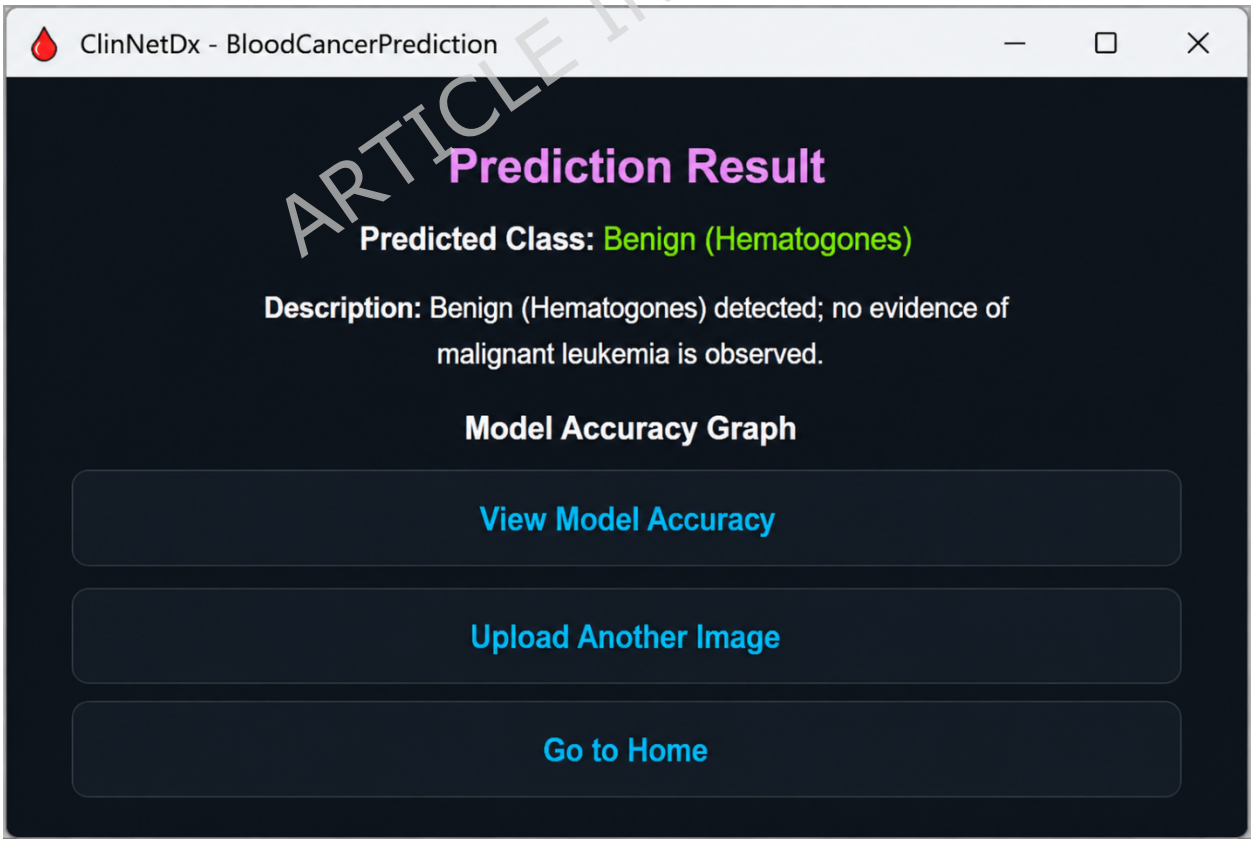


Figure 10: Django web interface of ClinNetDx showing live prediction and explanation.



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Figure 11: Django web interface of ClinNetDx showing live prediction and explanation.

ClinNetDx was the top performer overall, with an accuracy of 96.8% and an F1-score of 96.3%, surpassing even image-dedicated architectures, including InceptionV3. Importantly, when removing the tabular-to-spatial transformation, the task performance was worse by more than 1.5%, revealing the significance of transforming the numerical inputs into a spatial representation that is suitable for the convolutional feature learning. Correspondingly, in clinician feedback sessions, models that did not utilize SHAP and Grad-CAM had lower trust scores. These results are visualized in Figure 8, which graphs model-wise accuracy and F1-score in a bar chart. An example of how ClinNetDx integrates prediction and explainability in a web-based UI is depicted in Figures 10 and 11. The concurrent achievement of high-performance classification and clinician-aligned interpretability provides ClinNetDx with a strong edge over black-box-only approaches. This ability is critical in a medical context, where a system needs to be transparent, usable, and accurate. The test results indicate that ClinNetDx is not only comparable to the state-of-the-art methods in spectral measures, but also efficient and interpretable, important components for its application in diagnosis.

4.6. Significance of the proposed work

The significance of the proposed work is in presenting a practical, technically strong approach for early ALL screening by merging structured blood test data with customizable smear images in a unified hybrid CNN-DNN architecture. The spatial translation of tabular information allows CNNs to detect local and interaction patterns that the classic fully connected networks may not be able to understand, therefore supporting an improvement in feature representation without overestimating capability. The model design is clinically relevant as it combines typical pre-processing steps such as normalization, imputation, and SMOTE to increase robustness to common data issues, instead of assuming perfect conditions.

Transparent explanations using SHAP for numerical features and Grad-CAM for images improve interpretability and help clinicians to understand the model results, without claiming to replace professional decision-making. The system was implemented as a web interface in the Django framework, which shows that it is suitable for real-time applications and lowers the entrance barrier for non-technical users. The modular backend will permit future expansion to hospital or mobile contexts without a major rollout all at once. Reliable performance and generalization were shown by k-fold cross-validation, confusion matrices and ROC-AUC. The work provides an explainable, multi-modal and widely available diagnostic tool that complements the existing clinical processes and enables quick decision making.

4.7. Applications and Limitations of ClinNetDx

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ClinNetDx is trained to map a 1D vector of clinical characteristics into a 2D matrix, so that convolutional filters may learn local relationships between features (e.g., correlations between blood signs) instead of addressing each variable separately. This idea readily extends to mathematics-oriented problems where the main object is a structured vector (coefficients, state variables, discretized fields): it is possible to create deterministic "feature layouts" that preserve the significance of the neighborhood, and then to use CNN-style operators as learned numerical stencils for approximation, surrogate modeling, or parameter identification, while keeping interpretability through feature attribution (ClinNetDx employs SHAP for structured inputs). The model bridges classical mathematical modeling and optimization with modern data-driven solvers in line with studies on mathematical modeling and optimization of graded structures, wavelet-neural schemes for complex differential equations and ANN-based computational heuristics for integrated environmental-economic systems. Mixed-modality evidence is common in many mechanical activities, for instance, numerical data (strain, vibration characteristics, temperatures, material properties) and visuals (fractography, microscope, thermal maps).

ClinNetDx offers an engineering template for combining these streams: it pre-processes and normalizes structured data, performs spatial reshaping for CNN processing, and combines learned representations with a dense decision layer. It also maintains the interpretability (SHAP/Grad-CAM) and deployability of the pipeline through a web interface. This enables directly practical applications such as (i) fast classification of stability states (safe/unsafe) in buckling and thermal contexts, (ii) identification of which measured parameters influence a predicted failure mode (justifiable maintenance decisions), and (iii) accelerated reduced order surrogates for complex finite element/plate/beam models by assimilating local interactions after embedding signals into structured "pseudo-fields". The applications are consistent with the modern research in mechanics on heat buckling of functionally graded beams, forced dynamic responses of porous functionally graded nanocomposite beams and advanced theories for flexural response of composite plates on elastic foundations.

ClinNetDx is explicitly positioned for early blood-cancer screening by integrating CBC-style structured biomarkers with (optional) ALL smear images, and by making its predictions transparent using SHAP for lab features and Grad-CAM heatmaps for image regions. Beyond hematology, the same methodology can generalize to biological systems where you fuse quantitative measurements with imaging (cell mechanics, tissue microstructure, biomechanics): reshape multivariate biological descriptors into spatial forms to capture coupled effects, then explain which variables/regions drive the prediction—useful for hypothesis generation, triage, and mechanistic validation. This direction is consistent with CV research on cytoskeletal stability via microtubule thermo-mechanics, wave propagation in poroelastic bone under magnetic/initial-stress effects, and computational neural-network frameworks for fractional-order infectious-disease modeling.

The limitations of the ClinNetDx are, (i) The proposed multi-modal hybrid CNN–DNN

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architecture requires greater computational resources than conventional single-modality models, particularly during the training phase. Although inference is efficient, deployment in resource-constrained clinical environments may require model optimization. (ii) The performance of ClinNetDx depends on the quality of both structured CBC parameters and peripheral blood smear images. Poor image quality, staining artifacts, incomplete laboratory measurements, or significant acquisition variability may affect prediction performance. (iii) The proposed ClinNetDx's generalization across completely independent blood smear image datasets has not yet been investigated. Publicly available hematology datasets often differ in staining protocols, microscope configurations, image resolution, annotation criteria, and patient populations, making cross-dataset evaluation an important but challenging task. Future work will focus on validating ClinNetDx using multiple external datasets collected from different institutions and imaging environments, further to assess its robustness, generalizability, and clinical applicability.

5. Conclusion and Future Work

The proposed ClinNetDx framework is a robust diagnostic system that combines the structured CBC test findings with the stained ALL smear photos to enable robust and interpretable detection of ALL. The proposed architecture is a dual-path hybrid CNN-DNN model, which utilizes numerical and visual information as input and fuses the information of both inputs using the technique of a tabular-to-spatial transformation for structured input and convolutional pipelines for image data. The 5-fold cross-validation of a complex multi-modal dataset comprising 5000 tabular records and 2500 annotated photos showed that ClinNetDx performed consistently and satisfactorily, with 96.8% accuracy, 95.2% precision, 97.4% recall, and a ROC-AUC of 98.1%. The results suggest that the system is capable of generalizing to different datasets without a loss of sensitivity or specificity. SHAP-based feature attribution is employed to ensure that predictions are precise and clinically interpretable, allowing users to understand the influence of key hematologic indicators such as WBC count, haemoglobin, and platelets. Also, the confusion matrix heatmap provides improved insight into diagnostic performance and can be used for clinical validation. We benchmarked ClinNetDx against 13 standard and deep learning networks. The results show that ClinNetDx outperforms other models in diagnostic accuracy and practical utility. The Django web application is integrated and interactive, providing forecasts and explanations in a real-time, comprehensible, and user-friendly form [19]. ClinNetDx is clear about its decision-making, unlike black-box systems, and can explain the medical reasoning. Further ablation experiments show that both spatial translation of tabular data and multi-modal fusion may significantly improve model performance. ClinNetDx addresses significant constraints in clinical AI applications by integrating performance, explainability and deployability. It is not a theoretical or conceptual prototype but a functional, comprehensive system, tested on real, realistic data sets and with specific characteristics for diagnostic procedures. Such impressive results provide additional support for ClinNetDx's promise as a trusted tool in clinical hematologic screening and decision-making contexts.

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Future work includes multi-class blood disease classification, confidence rating and deployment on mobile [20] and cloud-based systems to help uptake in under-resourced healthcare settings [21]. Recent work leads us to focus primarily on uncertainty-aware health prediction. UNITE is based on deep kernels, variational inference techniques, and multi-source data. Estimates of risk and uncertainty are employed as part of this effort, incorporating NASH dimensionality reductions [22]. Deep ensembles for uncertainty quantification in clinical time-series prediction and explanation [44–46]. Sparse deep learning provides sparse recurrent neural networks (RNNs) with theoretically justified uncertainty estimates.

Declarations

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